

1. Administrative & Study Identification

Full Study Title: A Phase 2b/3, Randomized, Double-Blind Study to Investigate the Efficacy, Safety, and Tolerability of Ponsegromab (PF-06946860) Compared With Placebo Both With Background First-Line Chemotherapy in Adult Participants With Cachexia and Metastatic Pancreatic Ductal Adenocarcinoma **Short Title/Acronym:** RIVER-mPDAC / RIVER Pancreatic **Protocol Number:** C3651021 **NCT Number:** NCT06989437 **Sponsor Name:** Pfizer Inc. **Investigational Product:** Ponsegromab (PF-06946860) (GDF-15 Monoclonal Antibody) **Phase of Development:** Phase IIb/III **Medical Monitor:** Pfizer Clinical Trial Contact Center / Pfizer Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the efficacy, safety, and tolerability of ponsegromab compared with placebo, both administered with background first-line chemotherapy, in adult participants with cachexia and metastatic pancreatic ductal adenocarcinoma (mPDAC). **Secondary Objectives:** To evaluate the effect of ponsegromab on physical activity, Overall Survival (OS), patient-reported physical function, and fatigue levels, as well as to characterize the overall safety and tolerability profile (including adverse events and laboratory abnormalities).

2.2 Background & Rationale To address the critical unmet medical need for treating cachexia in advanced cancer patients. Cachexia—characterized by severe unintended weight loss, muscle loss, poor appetite, and extreme tiredness—is not merely a side effect of cancer treatment, but a separate condition driven by high levels of GDF-15, which alters the body's energy metabolism and speeds up calorie burning. By blocking GDF-15 with ponsegromab, this study aims to control cachexia symptoms, improve patients' ability to tolerate life-prolonging cancer treatments (like nab-paclitaxel + gemcitabine or FOLFIRINOX), and enhance their overall quality of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** * *Phase 2b:* Randomized 1:1:1 (ponsegromab 200 mg, ponsegromab 400 mg, or placebo), stratified by ECOG PS, chemotherapy regimen, and BMI-adjusted weight loss. * *Phase 3:* Randomized 1:1 (selected ponsegromab dose or placebo).

3.2 Planned Enrollment & Duration Target **N\$:** Approximately 1,000 adult participants **Number of Sites:** Global, multicenter study (including United States, Australia, Brazil, Canada, China, India, Israel, Japan, Puerto Rico, South Korea, Taiwan) **Estimated Study Start:** May-2025 **Estimated Primary Completion:** TBD (Active enrolling, no interruption between Phase 2b and 3)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Documented active diagnosis of metastatic pancreatic ductal adenocarcinoma (mPDAC) in adults (≥ 18 years). 2. Cachexia defined by Fearon criteria of weight loss. 3. Completed 1×28 -day cycle of first-line systemic nab-paclitaxel and gemcitabine chemotherapy or 2×14 -day cycles of FOLFIRINOX chemotherapy and prior to receiving the next cycle. 4. ECOG Performance Status ≤ 1 with a life expectancy of at least 4 months.

4.2 Exclusion Criteria 1. Current active reversible causes of decreased food intake, or cachexia caused by reasons other than cancer. 2. Any prior or current clinical diagnosis of heart failure, or Left ventricular ejection fraction $< 50\%$. 3. Receiving tube feedings or parenteral nutrition at the time of Screening or Randomization. 4. Neuroendocrine or acinar pancreatic carcinoma, symptomatic brain metastasis, leptomeningeal disease, or other active CNS metastases. 5. Inadequate liver function or renal disease requiring dialysis (eGFR < 30 mL/min/ 1.73m^2).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pongegromab (200 mg or 400 mg in Phase 2b; optimized dose in Phase 3) or matching placebo administered once every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Double-blind treatment period until disease progression or unacceptable toxicity, followed by an optional open-label extension period (≤ 1 year).

5.2 Concomitant & Prohibited Therapy Permitted: Background first-line chemotherapy (nab-paclitaxel + gemcitabine, or FOLFIRINOX) as determined by the investigator per local guidelines. **Prohibited:** Tube feedings or parenteral nutrition at screening/randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day -1 Informed Consent, Inclusion/Exclusion Assessment, ECOG PS, LVEF, Labs Baseline

| Day 1 | Randomization, First SC Injection (administered on the same day as next cycle of chemotherapy, prior to chemo), Baseline PROs | | **Interim** | Q4W | Subcutaneous injections of ponesegromab/placebo, Safety Labs, Weight measurements, FAACT-5IASS and PROMIS questionnaires | | **Primary Assessment** | Week 12 | Primary endpoint evaluation (Weight and Appetite changes), Dose selection for Phase 3 | | **End of Study** | TBD | Final safety assessments, OS Follow-up, Optional Open-Label Extension |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percent change from baseline in body weight at Week 12, AND Change from baseline in appetite-related symptoms at Week 12. **Measurement Method:** Clinical weight measurement and the Functional Assessment of Anorexia/Cachexia Therapy 5-item Anorexia Symptom Scale (FAACT-5IASS).

7.2 Secondary & Exploratory Endpoints * **Physical Activity:** Change from baseline in physical activity at Week 12. * **Overall Survival (OS):** Time from randomization to death due to any cause. * **Patient-Reported Outcomes:** Change in physical function and fatigue assessed via PROMIS Physical Function and PROMIS Fatigue questionnaires. * **Safety:** Incidence of Treatment Emergent Adverse Events (TEAEs), SAEs, laboratory abnormalities, and symptomatic AEs (NCI PRO CTCAE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of TEAEs, symptomatic AEs (assessed via NCI PRO CTCAE for diarrhea, nausea, vomiting, decreased appetite, fatigue, mouth sores), and lab/vital sign abnormalities. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent DMC will review unblinded safety and efficacy data, including evaluating the Phase 2b data to recommend the optimal ponesegromab dose for Phase 3.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) set for efficacy endpoints; Safety Analysis Set for evaluating AEs and tolerability. **Sample Size Justification:** Target enrollment of \$ \approx 1,000\$ participants across Phase 2b/3 ensures adequate statistical power to detect significant differences in body weight, appetite symptoms (Week 12), and overall survival (OS) between the ponesegromab and placebo arms. **Handling of Missing Data:** Appropriate imputation methods (e.g., Mixed-Model Repeated Measures [MMRM] or Multiple Imputation) will be applied for missing patient-reported outcomes and weight measurements.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Sponsor-directed onsite and remote monitoring to verify protocol compliance, source data verification, and accurate eCRF entry. **Audit Trail:** Secure, 21 CFR Part 11 compliant electronic data capture (EDC) system with full audit trails for data entry and query resolution.

11. Study Identifiers & Governance

Primary ID: C3651021 **Registry Number:** NCT06989437 **Sponsor/Lead Organization:** Pfizer Inc. **Study Title:** RIVER-mPDAC (RIVER Pancreatic) **Official Regulatory Title:** A Phase 2b/3, Randomized, Double-Blind Study to Investigate the Efficacy, Safety, and Tolerability of Ponsegromab (PF-06946860) Compared With Placebo Both With Background First-Line Chemotherapy in Adult Participants With Cachexia and Metastatic Pancreatic Ductal Adenocarcinoma.

12. Clinical Framework (Oncology/Cachexia Specific)

Primary Condition: Cachexia secondary to Metastatic Pancreatic Ductal Adenocarcinoma (mPDAC) **Diagnosis Threshold:** Cachexia defined by Fearon criteria of weight loss **Medical Methodology:** GDF-15 inhibition via monoclonal antibody (Ponsegromab) coupled with standard first-line systemic chemotherapy. **Optimization Target:** Weight stabilization/gain, appetite improvement, and functional symptom management.

13. Study Procedures & Methodology

Management Pathway: Integration of subcutaneous ponsegromab/placebo Q4W seamlessly into standard-of-care chemotherapy infusion cycles. **Education Protocol (HCP):** Site training on Investigational Product (IP) handling, SC injection techniques, and administration of PRO instruments. **Education Protocol (Patient):** Guidance on recognizing cachexia symptoms, tracking physical activity, and completing digital FAACT-5IASS and PROMIS questionnaires. **Quality Audit Mechanism:** Tracking of timely PRO completion and ongoing chemotherapy regimen adherence.

14. Observation & Follow-Up Schedule

Total Duration: Double-blind period + optional open-label extension (≤ 1\$ year) + long-term Overall Survival (OS)

follow-up. **Onsite Visit Cadence:** Every 4 Weeks (Q4W) matching chemotherapy cycles. **Remote Monitoring Frequency:** Remote safety follow-ups as determined by the clinical protocol. **Checklist Review Interval:** Regular evaluations with a major primary efficacy data lock occurring after the Week 12 milestone.

15. Sign-off & Approval

Role	Name	Signature	Date	_____	_____	_____	_____
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				